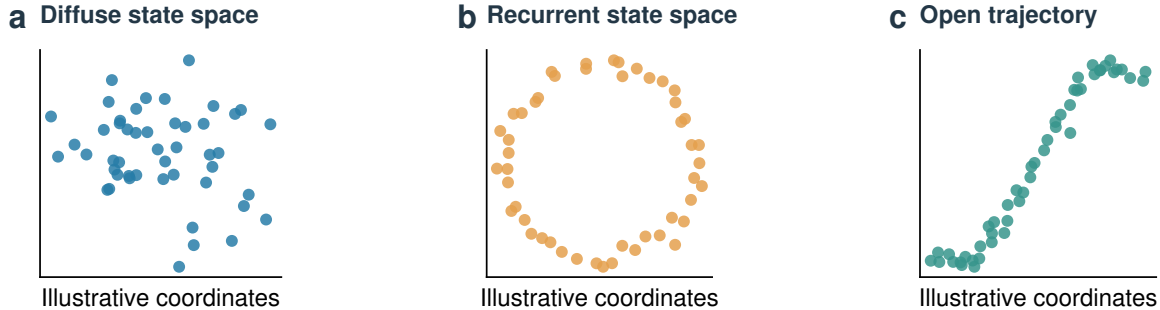
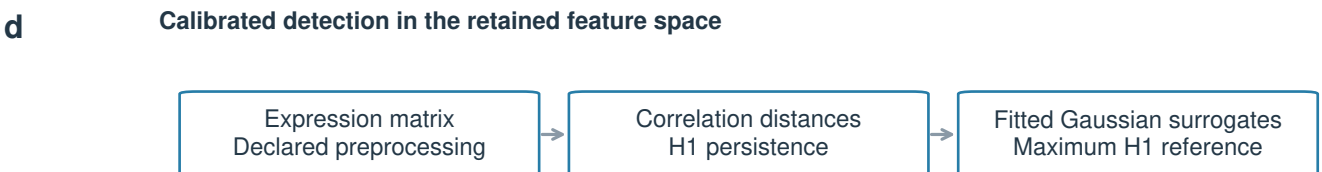
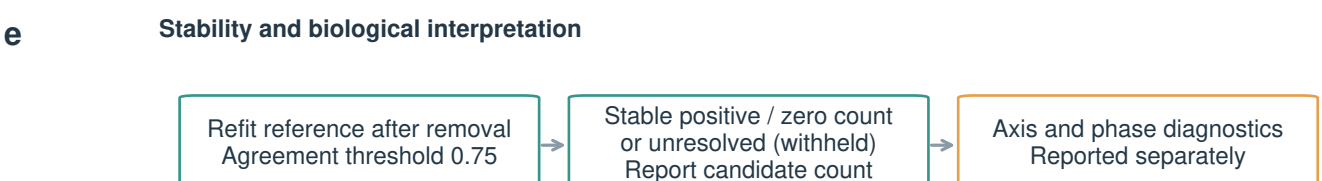
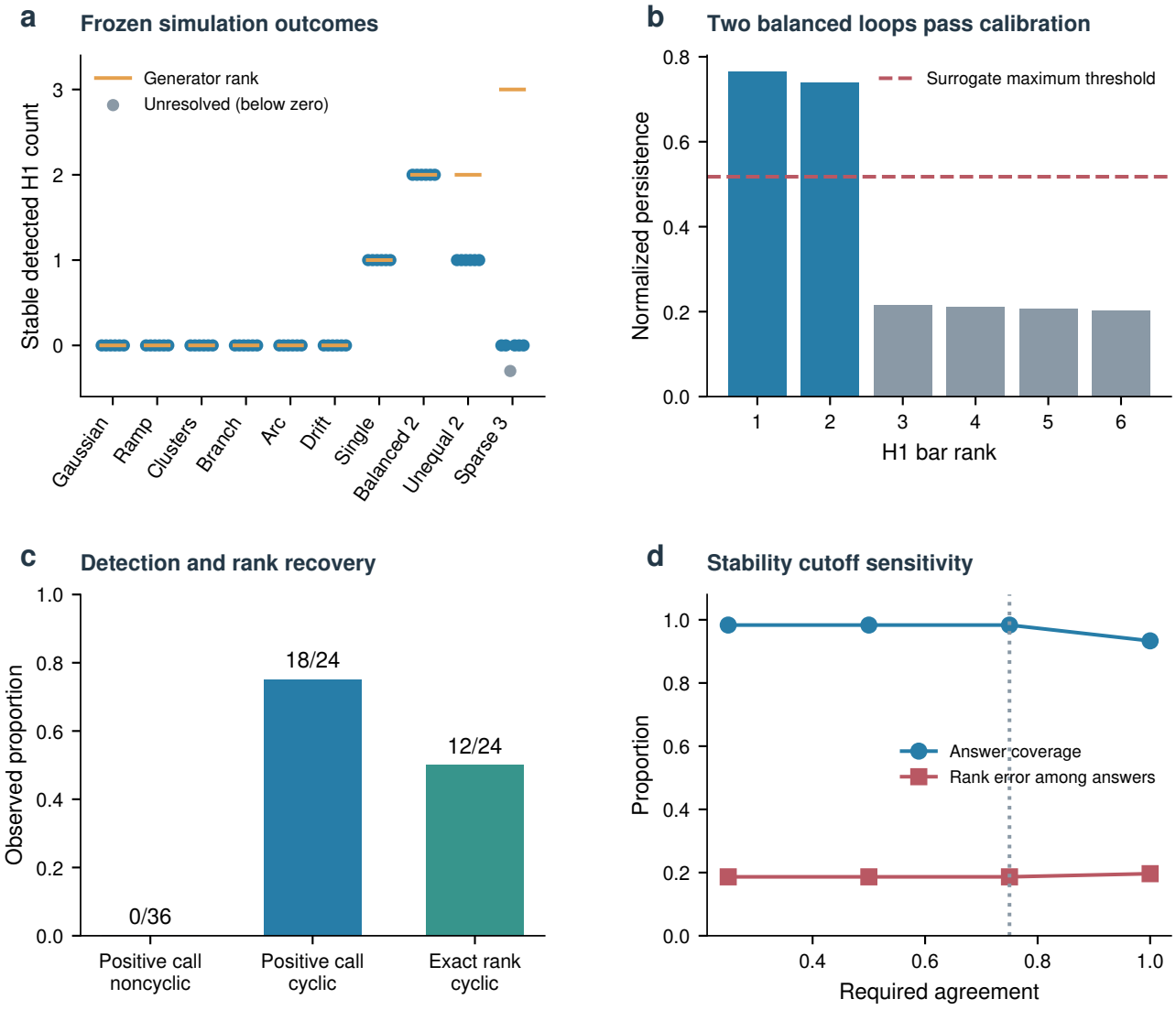
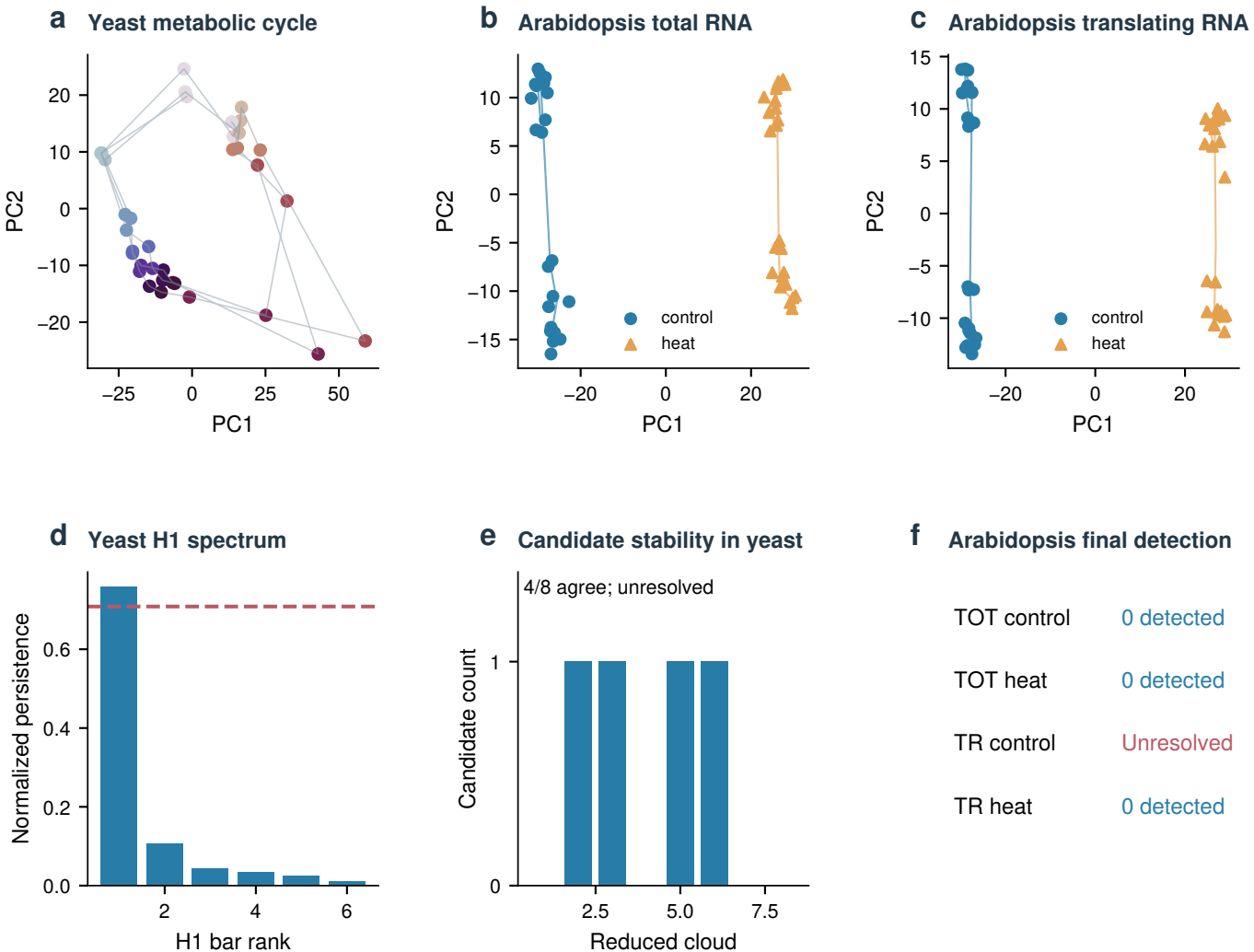
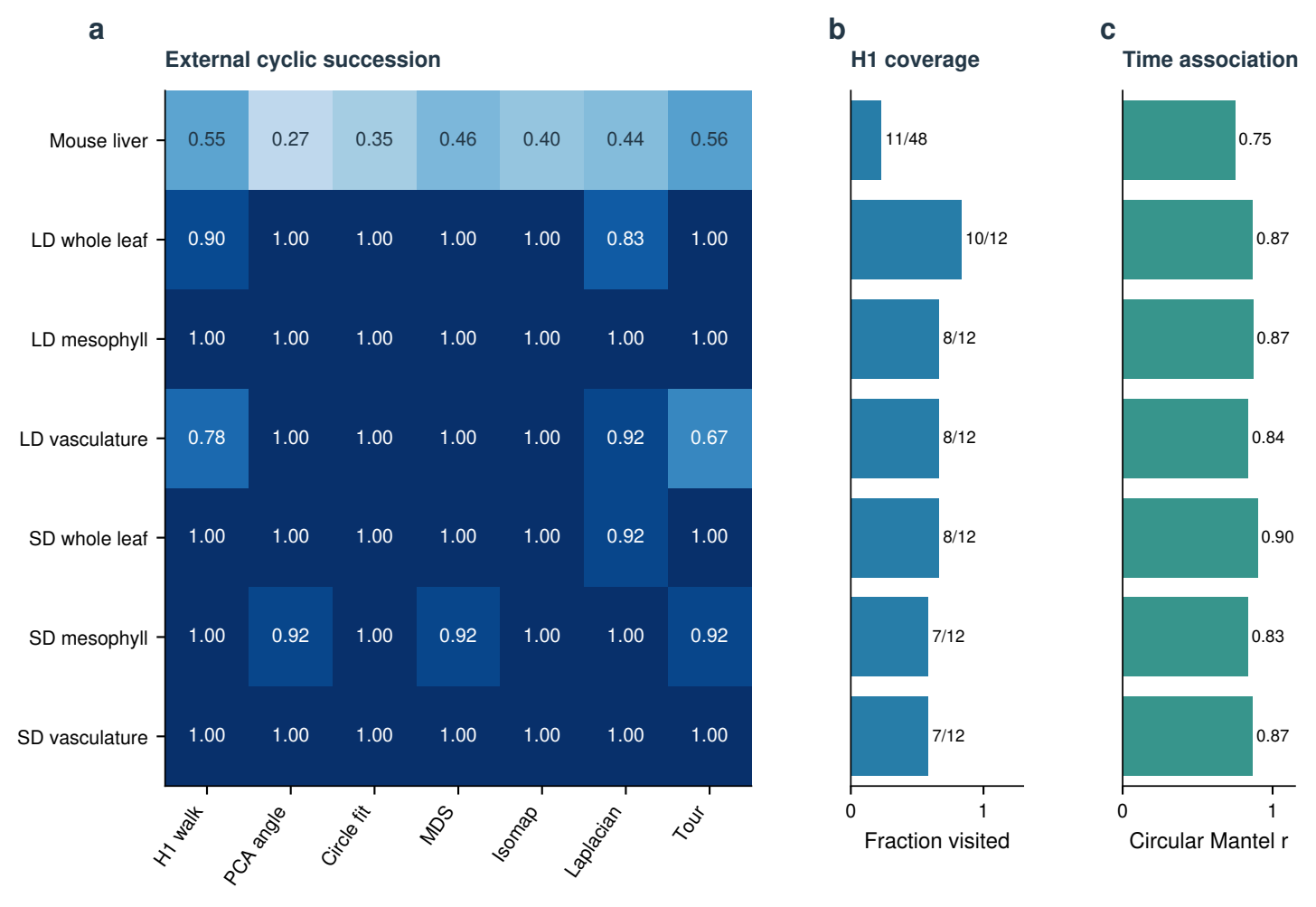

a Diffuse state space
Illustrative coordinates
b Recurrent state space
Illustrative coordinates
c Open trajectory
Illustrative coordinates

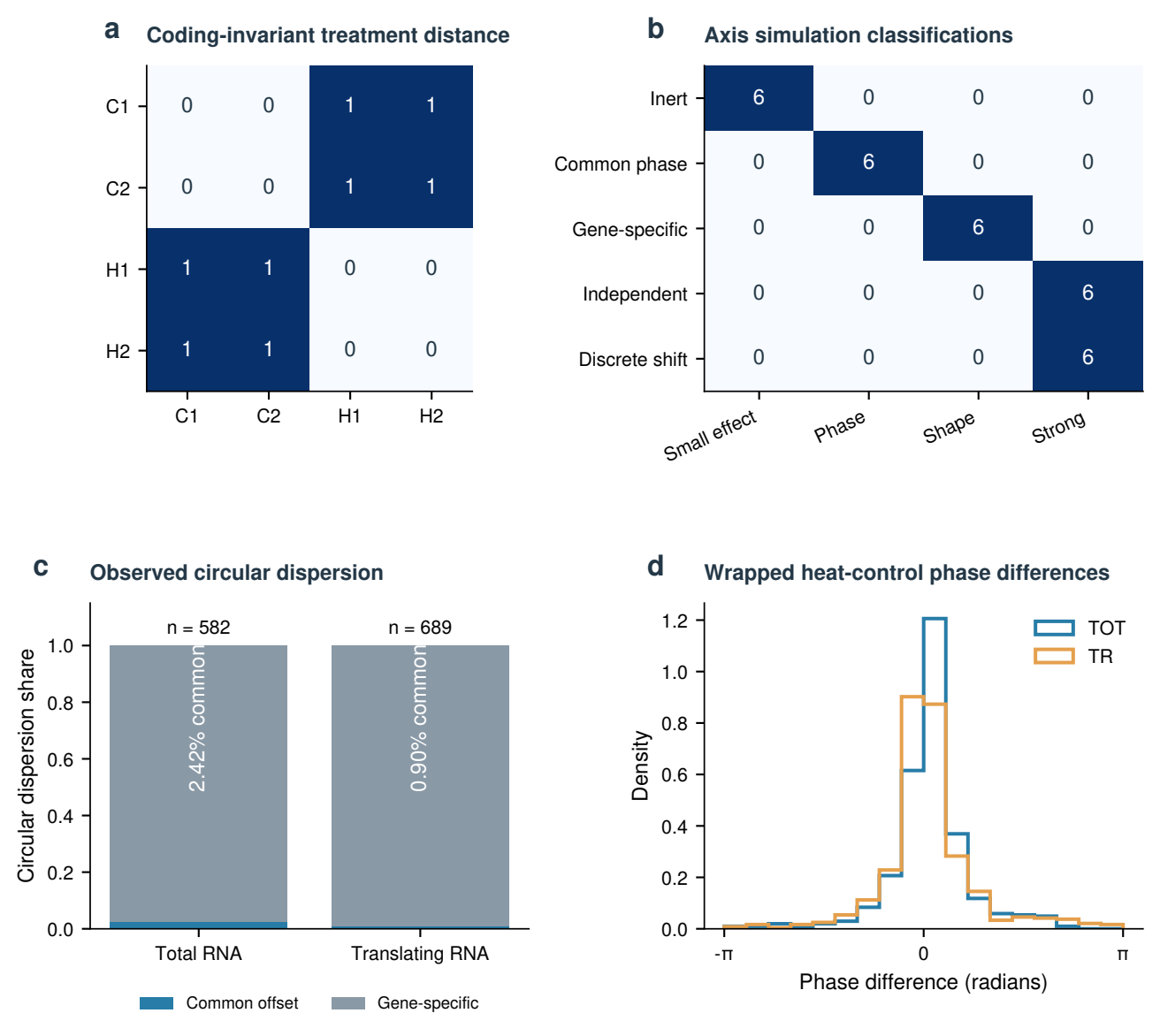

d
Calibrated detection in the retained feature space
Expression matrix
Declared preprocessing
Correlation distances
H1 persistence
Fitted Gaussian surrogates
Maximum H1 reference

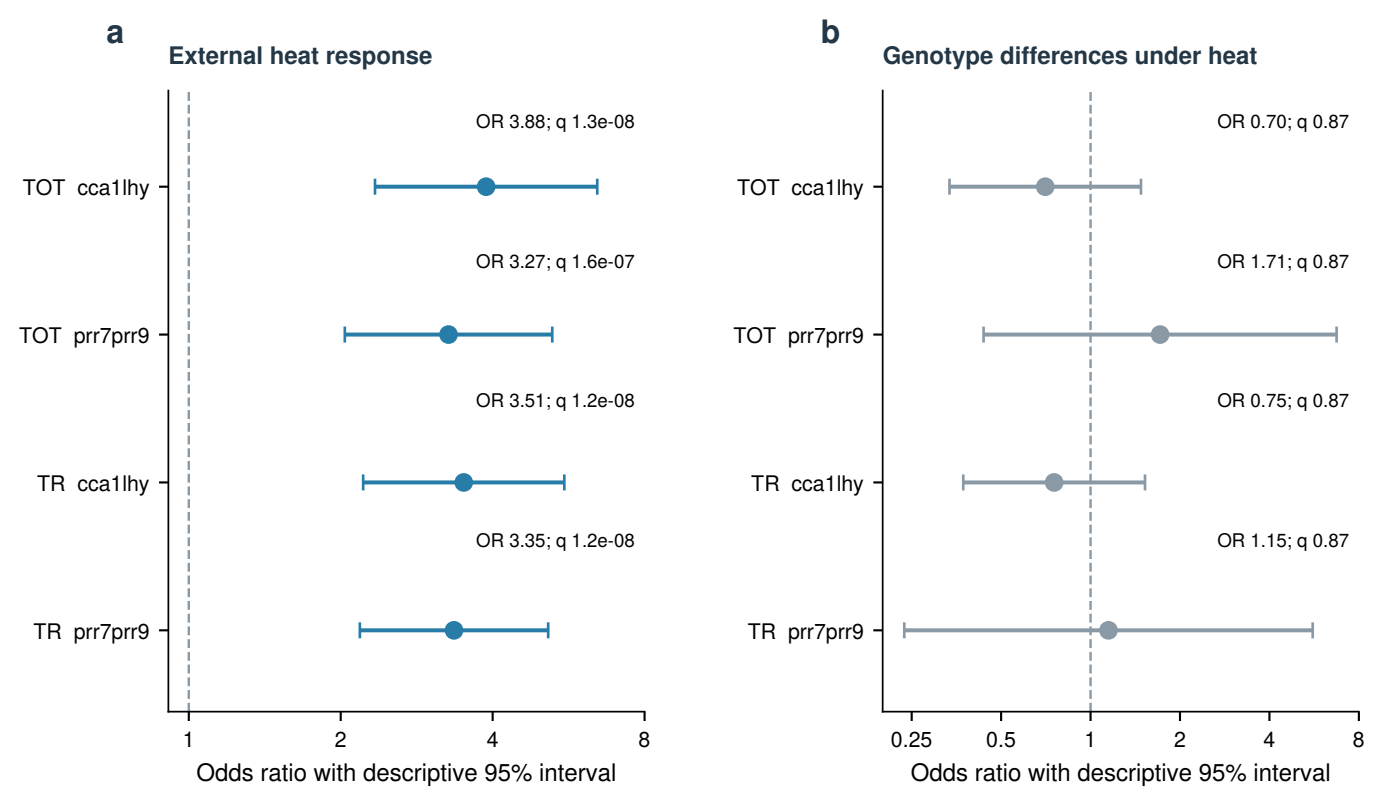

e
Stability and biological interpretation
Refit reference after removal
Agreement threshold 0.75
Stable positive / zero count
or unresolved (withheld)
Report candidate count
Axis and phase diagnostics
Reported separately


a
Frozen simulation outcomes
Generator rank
Unresolved (below zero)
Stable detected H1 count
0
1
2
3
Gaussian
Ramp
Clusters
Branch
Arc
Drift
Single
Balanced 2
Unequal 2
Sparse 3
b
Two balanced loops pass calibration
Surrogate maximum threshold
Normalized persistence
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
6
H1 bar rank
c
Detection and rank recovery
Observed proportion
0.0
0.2
0.4
0.6
0.8
1.0
0/36
18/24
12/24
Positive call noncyclic
Positive call cyclic
Exact rank cyclic
d
Stability cutoff sensitivity
Proportion
Answer coverage
Rank error among answers
0.4
0.6
0.8
1.0
Required agreement


a
Yeast metabolic cycle
PC2
PC1
b
Arabidopsis total RNA
control
heat
c
Arabidopsis translating RNA
d
Yeast H1 spectrum
Normalized persistence
H1 bar rank
e
Candidate stability in yeast
4/8 agree; unresolved
Candidate count
Reduced cloud
f
Arabidopsis final detection
TOT control 0 detected
TOT heat 0 detected
TR control Unresolved
TR heat 0 detected


a
External cyclic succession
Mouse liver
LD whole leaf
LD mesophyll
LD vasculature
SD whole leaf
SD mesophyll
SD vasculature
0.55 0.27 0.35 0.46 0.40 0.44 0.56
0.90 1.00 1.00 1.00 1.00 0.83 1.00
1.00 1.00 1.00 1.00 1.00 1.00 1.00
0.78 1.00 1.00 1.00 1.00 0.92 0.67
1.00 1.00 1.00 1.00 1.00 0.92 1.00
1.00 0.92 1.00 0.92 1.00 1.00 0.92
1.00 1.00 1.00 1.00 1.00 1.00 1.00
H1 walk
PCA angle
Circle fit
MDS
Isomap
Laplacian
Tour
b
H1 coverage
11/48
10/12
8/12
8/12
8/12
7/12
7/12
0
1
Fraction visited
c
Time association
0.75
0.87
0.87
0.84
0.90
0.83
0.87
0
1
Circular Mantel r


a
Coding-invariant treatment distance
C1
C2
H1
H2
0 0 1 1
0 0 1 1
1 1 0 0
1 1 0 0
C1 C2 H1 H2
b
Axis simulation classifications
Inert
Common phase
Gene-specific
Independent
Discrete shift
6 0 0 0
0 6 0 0
0 0 6 0
0 0 0 6
0 0 0 6
Small effect
Phase
Shape
Strong
c
Observed circular dispersion
n = 582
n = 689
2.42% common
0.90% common
Circular dispersion share
1.0
0.8
0.6
0.4
0.2
0.0
Total RNA
Translating RNA
Common offset
Gene-specific
d
Wrapped heat-control phase differences
TOT
TR
Density
1.2
1.0
0.8
0.6
0.4
0.2
0.0
-π
0
π
Phase difference (radians)


a
External heat response
OR 3.88; q 1.3e-08
TOT cca1lhy
OR 3.27; q 1.6e-07
TOT prr7prr9
OR 3.51; q 1.2e-08
TR cca1lhy
OR 3.35; q 1.2e-08
TR prr7prr9
1
2
4
8
Odds ratio with descriptive 95% interval
b
Genotype differences under heat
OR 0.70; q 0.87
TOT cca1lhy
OR 1.71; q 0.87
TOT prr7prr9
OR 0.75; q 0.87
TR cca1lhy
OR 1.15; q 0.87
TR prr7prr9
0.25
0.5
1
2
4
8
Odds ratio with descriptive 95% interval